FCM 1: Assignment 1

Dr. Gallivan

Alex Sampson

Contents

1 Executive Summary **3**

2 Statement of the Problem **3**

3 Mathematical Description **4**

4 Implemented Subroutines **7**

5 Driver and Experiments **8**

6 Results and Discussion **8**

7 Set I: **8**

8 Set II: **10**

9 Conclusions **12**

10 Disclaimer **12**

A Figures **13**

1 Executive Summary

I implemented in-place LU factorization with selectable pivoting (`mode =0 none, 1 partial, 2 complete`), along with forward/backward triangular solvers and small supporting routines to apply permutations and multiply stored factors. I then ran trend studies across problem sizes and matrix classes to evaluate:

- Relative factorization error: $\dfrac{\|P_r A P_c - LU\|}{\max(1, \|A\|)}$,
- Relative solution error: $\dfrac{\|x_{\text{true}} - \hat{x}\|}{\max(1, \|x_{\text{true}}\|)}$
- Residual: $\dfrac{\|b - A\hat{x}\|}{\max(1, \|b\|)}$,
- Growth factor: $\gamma = \dfrac{\|\,|L|\,|U|\,\|}{\|A\|}$

These were evaluated on random full-rank matrices, partial and complete pivoting behave stably, while no-pivoting shows occasional failures or larger γ. Figures in the Results section summarize errors vs. n and the γ distributions by mode.

2 Statement of the Problem

We implement LU factorization for $A \in \mathbb{R}^{n\times n}$ via a storage array M while using Gaussian elimination with selectable pivoting so that

$$P_r\, A\, P_c \;=\; L\, U,$$

where L is unit-lower triangular, U is upper triangular, and P_r, P_c are permutation matrices represented by integer index arrays `pr`, `pc`. The pivoting modes are:

(i) no pivoting,

(ii) partial (row) pivoting,

(iii) complete pivoting.

Near-zero pivots are detected using a tolerance `tol`, the routines identify these and produce errors/warning

Solving $Ax = b$ once $P_r A P_c = LU$ is known:

Given a right-hand side b, the computed solution $\hat{x}$ is obtained by

$$\begin{aligned}
b_r &= P_r b, \quad \rightarrow && \text{computed directly}\\
Ly &= b_r, \quad \rightarrow && \text{solved using forward substitution}\\
Uz &= y, \quad \rightarrow && \text{solved using backward solve}\\
\hat{x} &= P_c z.
\end{aligned}$$

For partial pivoting $P_c = I$, so $\hat{x} = z$. We assess the solve with

$$\text{relX} = \frac{\|x_{\text{true}} - \hat{x}\|}{\max(1, \|x_{\text{true}}\|)}, \qquad \text{relR} = \frac{\|b - A\hat{x}\|}{\max(1, \|b\|)}.$$

What we compute and report:

For a chosen matrix norm, we record:

$$\text{relF} = \frac{\|P_r A P_c - LU\|}{\max(1, \|A\|)}, \qquad \gamma = \frac{\big\| \, |L| \, |U| \, \big\|}{\|A\|},$$

together with relX, relR when a solve is performed. (Here $|\cdot|$ is element-wise absolute value).

Set 1: Evaluation 8 on small structured matrices

We run a fixed collection of eight small 8×8 examples as outlined in the program guidelines. For each example and for each pivoting mode,

- we form $P_r A P_c = LU$ and verify relF is near rounding error when no breakdown occurs.
- we solve $Ax = b$ for a simple b (e.g., $b = \mathbf{1}$ or a random vector) and report relX, relR;
- we compute γ to illustrate element growth under each pivoting rule.

Set 2: Trend study on random dense matrices

We study scaling behavior over a range of sizes n (e.g., 30:10:250) and multiple trials per n for representative distributions. For each n, distribution, and pivoting mode, we

- compute relF, relX, relR, γ for each trial,
- aggregate mean and max values versus n.
- visualize γ via histograms/densities by pivoting mode to compare stability profiles.

This set highlights how the metrics trend with problem size and how pivoting strategies impact numerical stability and solve accuracy in practice.

3 Mathematical Description

The single storage array M stores L and U. As usual we consider the diagonal 1's of L in M implicit.

Permutations: Integer arrays `pr`, `pc` represent the row and column permutation matrices P_r, P_c respectively. To apply them to any matrix B we use MATLAB indexing $P_r B P_C$ =`B(pr,pc)`. We also explicitly developed a subroutine `apply_perms.m` as requested by the program outline

Factorization error: With $A_{\text{perm}} = P_r A P_c$ and factors from M, the relative error is

$$\frac{\|A_{\text{perm}} - LU\|}{\max(1, \|A\|)}.$$

Solve: Once we have successfully computed the LU factorization of A, solving $LUx = b$ consists of two triangular solves: $Ly = b$ using `forward_unit.m` and $Uz = y$ using `back_sub.m` as well as applying permutations `pr` and `pc` appropriately.

Growth factor: We form $|L|\,|U|$ directly from M. Since the diagonal of L is implicit (all ones) in M, there we simply add a mode in `LUmult_outer.m` from Program 1 to preform the multiplication $|L||U|$. Here,

$$\gamma \;=\; \frac{\big\|\,|L|\,|U|\,\big\|}{\|A\|}.$$

Pivoting choices: We want to explore several pivoting strategies in order to successfully perform the gaussian elimination process. To do so we develped multiple modes in our routines. These modes are designated `mode=0,1,2` and select none, partial or complete pivot rules respectively. Any near zero pivots are detected by our code in order to avoid any breaks.

Worked Examples

As usual we use a single storage array M for both factors L and U where L is unit lower triangular and the diagonal 1's are stored implicitly.

Example 1 partial pivoting: Suppose

$$A = \begin{bmatrix} 2 & \frac{9}{2} & 2 & -2 \\ -4 & \frac{13}{3} & \frac{5}{3} & \frac{14}{3} \\ 4 & -1 & 2 & 0 \\ 6 & \frac{-13}{2} & 3 & 3 \end{bmatrix}.$$

Initialize: `pr`=[1,2,3,4], `pc`=[1,2,3,4].

Step 1 (column 1): The largest value in the pivot column is in row 4 so we swwap rows $1 \leftrightarrow 4$. Therfor `pr` $\leftarrow$ [4,2,3,1].

New multipliers: $\ell_{2,1} = \frac{1}{3},\ \ell_{3,1} = -\frac{2}{3},\ \ell_{4,1} = \frac{2}{3}$.

Updated U:

$$\begin{bmatrix} 6 & \frac{-13}{2} & 3 & 3 \\ 0 & 0 & \frac{11}{3} & \frac{20}{3} \\ 0 & \frac{10}{3} & 0 & -2 \\ 0 & \frac{20}{3} & 1 & -3 \end{bmatrix}.$$

Step 2 (column 2): Similarly, swap rows $2 \leftrightarrow 4$. `pr` $\leftarrow$ [4,1,3,2].

New multipliers: $\ell_{3,2} = \frac{1}{2},\ \ell_{4,2} = 0$.

Updated U:

$$\begin{bmatrix} 6 & \frac{-13}{2} & 3 & 3 \\ 0 & \frac{20}{3} & 1 & -3 \\ 0 & 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{11}{3} & \frac{20}{3} \end{bmatrix}.$$

Step 3 (column 3): Swap rows $3 \leftrightarrow 4$. `pr` $\leftarrow$ [4,1,2,3].
New multiplier: $\ell_{4,3} = -\frac{3}{22}$.
Updated U:

$$\begin{bmatrix} 6 & \frac{-13}{2} & 3 & 3 \\ 0 & \frac{20}{3} & 1 & -3 \\ 0 & 0 & \frac{11}{3} & \frac{20}{3} \\ 0 & 0 & 0 & \frac{9}{22} \end{bmatrix}.$$

Step 4 (column 4): Pivot already on the diagonal, so `pr` $\leftarrow$ [4,1,2,3].

Finaly we have,

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{3} & 1 & 0 & 0 \\ -\frac{2}{3} & 0 & 1 & 0 \\ \frac{2}{3} & \frac{1}{2} & -\frac{3}{22} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 6 & \frac{-13}{2} & 3 & 3 \\ 0 & \frac{20}{3} & 1 & -3 \\ 0 & 0 & \frac{11}{3} & \frac{20}{3} \\ 0 & 0 & 0 & \frac{9}{22} \end{bmatrix}, \quad \texttt{pr} = [4,1,2,3], \quad \texttt{pc} = [1,2,3,4].$$

Example 2 complete pivoting: Suppose

$$A = \begin{bmatrix} \frac{23}{3} & \frac{-7}{3} & \frac{-5}{3} & \frac{8}{3} \\ \frac{-15}{2} & \frac{27}{2} & 10 & \frac{5}{2} \\ 10 & \frac{-27}{2} & \frac{5}{4} & \frac{3}{4} \\ -2 & 4 & 5 & 1 \end{bmatrix}.$$

Initialize `pr`=[1,2,3,4], `pc`=[1,2,3,4].

Step 1 pivot on submatrix: Swap rows $1 \leftrightarrow 2$ and cols $1 \leftrightarrow 2$, yielding `pr` $\leftarrow$ [2,1,3,4], and `pc` $\leftarrow$ [2,1,3,4].

We get the new multipliers: $\ell_{2,1} = -\frac{14}{81},\ \ell_{3,1} = -1,\ \ell_{4,1} = \frac{8}{27}$.

Our updated U is then:

$$U = \begin{bmatrix} \frac{27}{2} & \frac{-15}{2} & 10 & \frac{5}{2} \\ 0 & \frac{172}{27} & \frac{5}{81} & \frac{251}{81} \\ 0 & \frac{5}{2} & \frac{45}{4} & \frac{13}{4} \\ 0 & \frac{2}{9} & \frac{55}{27} & \frac{7}{27} \end{bmatrix}.$$

Step 2: Swap rows $2 \leftrightarrow 3$ and cols $2 \leftrightarrow 3$, yielding: `pr` $\leftarrow$ [2,3,1,4], `pc` $\leftarrow$ [2,3,1,4].

New multipliers: $\ell_{3,2} = \frac{4}{729}$, $\ell_{4,2} = \frac{44}{243}$.

So

$$U = \begin{bmatrix} \frac{27}{2} & 10 & \frac{-15}{2} & \frac{5}{2} \\ 0 & \frac{45}{4} & \frac{5}{2} & \frac{13}{4} \\ 0 & 0 & \frac{4634}{729} & \frac{2246}{729} \\ 0 & 0 & \frac{4225}{2187} & \frac{361}{2187} \end{bmatrix}.$$

Step 3: No swaps. So, `pr` $\leftarrow$ [2,3,1,4], `pc` $\leftarrow$ [2,3,1,4].

New multiplier: $\ell_{4,3} = -\frac{12}{331}$.

$$U = \begin{bmatrix} \frac{27}{2} & 10 & \frac{-15}{2} & \frac{5}{2} \\ 0 & \frac{45}{4} & \frac{5}{2} & \frac{13}{4} \\ 0 & 0 & \frac{4634}{729} & \frac{2246}{729} \\ 0 & 0 & 0 & \frac{-72}{331} \end{bmatrix}.$$

Step 4: No swaps and no multipliers. So our final LU factorization is:

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -\frac{14}{81} & \frac{4}{729} & 1 & 0 \\ \frac{8}{27} & \frac{44}{243} & -\frac{12}{331} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} \frac{27}{2} & 10 & \frac{-15}{2} & \frac{5}{2} \\ 0 & \frac{45}{4} & \frac{5}{2} & \frac{13}{4} \\ 0 & 0 & \frac{4634}{729} & \frac{2246}{729} \\ 0 & 0 & 0 & \frac{-72}{331} \end{bmatrix}, \quad \texttt{pr} = [2,3,1,4], \quad \texttt{pc} = [2,3,1,4].$$

4 Implemented Subroutines

- `my_lu(M,mode,tol)`: in-place LU with none, partial and complete pivoting, returns M, `pr`, `pc`, and status/warnings.
- `forward_unit(M,b)`: solve $Ly = b$ for unit-lower L stored in M.
- `back_sub(M,y,tol)`: solve $Ux = y$ for upper U stored in M.
- `apply_perms(A,pr,pc)`: return $P_r A P_c$ via indexing.
- `LUmult_outer(M,n,mode)`: multiply LU (or $|L||U|$) from packed M (outer-product scheme from project 1 but modified for $|L||U|$ from M).
- `growth_factor(M,A0,normtype)`: compute $\gamma = \dfrac{\| \, |L||U| \, \|}{\|A\|}$.
- `lu_solve(A,b,mode,tol)`: subroutine that calls `my_lu.m` and then solves the factored matrix M using `forward_unit.m` and `back_sub.m` that factors and solves.

5 Driver and Experiments

I run two sets of experiments:

Set I. Tests 8 'small' structured matrices and illustrates properties of L, U, P_r and P_c as well as the growth factor γ.

Set II. Generate a large range of n over larger random problems. The Set 2 driver sweeps n from 30 to 250 in steps of 10, repeats each case 5 times per class, and overlays pivoting modes. It aggregates means and maxima for each metric and builds a γ histogram per mode.

Matrix Classes for Set II

I use the following four representative classes:

(a) $A = L\,U$ with modest off-diagonals (well-behaved full rank),

(b) SPD $A = \tilde{L}\,\tilde{L}^\top$,

(c) strictly row-diagonally dominant,

(d) a permuted diagonally dominant matrix to obscure structure.

Metrics:

$$\text{relF} = \frac{\|P_r A P_c - LU\|}{\max(1, \|A\|)}, \quad \text{relX} = \frac{\|x_{\text{true}} - \hat{x}\|}{\max(1, \|x_{\text{true}}\|)}, \quad \text{relR} = \frac{\|b - A\hat{x}\|}{\max(1, \|b\|)}, \quad \gamma = \frac{\|\,|L|\,|U|\,\|}{\|A\|}.$$

6 Results and Discussion

7 Set I:

Below are qualitative observations about the packed factors L, U, the permutations P_r, P_c, and the growth factor γ for the eight structured 8×8 cases. For partial pivoting, note $P_c = I$ by design, complete pivoting changes both rows and columns.

To visualize these results I ran the following in the command window of MATLAB.

```
for i=1:8
    for j=0:2
        driver_set1(8,i,j,1e-12,1)
    end
end
```

(1) Diagonal (positive).

- *Structure:* Already upper triangular.
- *Factors:* $L = I$, $U = A$ under all modes. $P_r = P_c = I$ for none/partial, complete pivoting may reorder by magnitude but still produces $L = I$, U diagonal.
- *Growth:* $\gamma = 1$.

(2) Anti-diagonal (positive).

- *Structure:* Zeros on the main diagonal, all mass on the anti-diagonal.
- *Factors:* **No pivoting fails** immediately. Partial pivoting flips the rows so the large entries land on the diagonal (P_r reverses, $P_c = I$). complete pivoting can achieve the same with possible column swaps.
- *Growth:* After pivoting, $\gamma = 1$. The failure case has no meaningful γ.

(3) Diagonal + anti-diagonal.

- *Structure:* Diagonal with sizeable corner/anti-diagonal entries, invites element growth if not pivoted.
- *Factors:* Without pivoting, U shows large updated corner entries and visible fill pivoting move the large entries to safer pivot locations.
- *Growth:* None: γ noticeably larger, with pivoting: γ drops to ≈ 1.

(4) Unit lower with strictly subunit off-diagonals ($|\ell_{ij}| < 1$).

- *Structure:* Lower triangular, well conditioned for elimination.
- *Factors:* $L = A$, $U = I$. $P_r = P_c = I$ for none/partial complete pivoting does not need to move anything.
- *Growth:* $\gamma = 1$ across modes.

(5) Lower with diagonal > 0 and subdiagonal entries > 1.

- *Structure:* Lower triangular with larger subdiagonals, still safe.
- *Factors:* Elimination proceeds stably, partial/complete pivoting may perform benign row/column interchanges but do not change the qualitative pattern (lower L, upper U).
- *Growth:* $\gamma = 1$ in all modes in these runs.

(6) Tridiagonal, strictly diagonally dominant.

- *Structure:* Narrow band and strong diagonal dominance.
- *Factors:* Band is essentially preserved: L has a single subdiagonal of multipliers, U has a single superdiagonal, $P_r = P_c = I$ for none/partial.
- *Growth:* $\gamma \approx 1$ (slightly above 1 under complete pivoting in one run).

(7) Ones on last column, unit diagonal, -1 below diagonal.

- *Structure:* Each step pushes entries into the last column.
- *Factors:* None/partial pivoting leave the "accumulating" last column in place, so U entries escalate stage by stage. Complete pivoting swaps that heavy column into earlier positions (P_c nontrivial), dramatically reducing growth.
- *Growth:* None/partial: large γ (e.g., $\sim 6.3 \times 10^1$), complete: γ drops to a small value (e.g., ~ 2.75).

(8) SPD from $\tilde{L}\tilde{L}^\top$.

- *Structure:* Symmetric positive definite.
- *Factors:* Gaussian elimination is stable, partial pivoting does not need to move much. Complete pivoting may reorder by size without harming symmetry of the *permuted* system.
- *Growth:* $\gamma = 1$ across modes in these experiments.

Summary (Set 1).

- When the structure already presents good pivots (triangular, strictly diagonally dominant, SPD), P_r, P_c are typically the identity for none/partial, L, U mirror the input's triangular/banded form. $\gamma \approx 1$.
- When the structure hides good pivots (anti-diagonal, heavy last column), pivoting exposes them. Partial pivoting fixes row issues, complete pivoting additionally fixes column issues. This is visible directly in P_r, P_c and in the much smaller γ.
- The only outright failure here is "no pivoting" on the anti-diagonal matrix, which confirms why pivoting is essential in the presence of zeros or tiny diagonals.

8 Set II:

We report four metrics: relF, relX, relR, and the growth factor $\gamma = \| \, |L||U| \, \| / \|A\|$. Unless otherwise noted, curves are aggregated over multiple trials per n with modes overlaid.

Overall behavior:

For partial and complete pivoting (`mode = 1,2`), the errors remain essentially flat across n:

- **relF, relR:** near round-off over the full sweep (Figures 1, 3, 5, 7).
- **relX:** typically near round-off as well, occasional spikes align with difficult draws.
- **γ:** means are modest and distributions are close for modes 1–2 (Figures 2, 4, 6, 8).

No-pivoting (`mode 0`), when included, shows wider spread and occasional outliers in both relX and γ.

Classes:

Class 1 (product $A = LU$): Figures 1–2. relF/relR stay at machine precision with modes 1–2. γ is small on average with a light right tail.

Class 2 (Symmetric Positive Def): Figures 3–4. Similar to Class 1. Even without exploiting symmetry, pivoting yields uniformly small errors. γ histograms are concentrated near lower values.

Class 3 (strict row-diagonal dominance): Figures 5–6. Elimination is well behaved. Partial pivoting rarely needs large row moves. relF/relR flat, γ remains controlled. No-pivoting can be acceptable but is less predictable (broader tail).

Class 4 (permuted row-diagonal dominance): Figures 7–8. The permutation hides the diagonal advantage. Pivoting recovers good pivots: relF/relR remain small and γ is contained. In Figure 8, the histogram y-axis is a probability density (PDF), so areas normalize to 1.

Small residual but large solution error

We observed trials where the residual was tiny while the solution error was large. This is expected when the matrix is ill-conditioned.

Forward and Backward Error's: The residual (backward error) measures how well the computed $\hat{x}$ satisfies $A\hat{x} \approx b$. If

$$\frac{\|b - A\hat{x}\|}{\|b\|} \text{ is small,}$$

the algorithm is backward stable. The solution (forward) error measures how far $\hat{x}$ is from the true x:

$$\frac{\|x - \hat{x}\|}{\|x\|}.$$

Why the solution error can still be large: These two are linked by the condition number $\kappa(A)$:

$$\frac{\|x - \hat{x}\|}{\|x\|} \approx \kappa(A) \frac{\|b - A\hat{x}\|}{\|b\|}.$$

If $\kappa(A)$ is very large, even a very small residual can be amplified into a large solution error. This matches our maxima from the command window: relR $\approx 6.22 \times 10^{-11}$ (small) but relX $\approx 1.86 \times 10^{2}$ (large).

About the growth factor: A large γ indicates substantial element growth during elimination. Our highest γ was a rare event, for partial and complete pivoting the *average* and most of the distribution stay low (see Figures 2, 4, 6, 8).

Takeaways

- Partial/complete pivoting keep relF, relX, relR at round-off across sizes and classes.
- Growth factors under pivoting are modest on average with tight distributions, large γ is rare.
- No-pivoting is fragile: wider spreads and occasional large forward-error events.

9 Conclusions

Across the structured (Set 1) and random (Set 2) tests, the behavior matches the standard stability picture for Gaussian elimination. When structure already supplies good pivots (triangular, strictly diagonally dominant, SPD), the factors L, U preserve the expected triangular/banded form, P_r, P_c are largely identity permutation, and the growth factor $\gamma \approx 1$. On more difficult structures (anti-diagonal or "heavy last column"), pivoting is essential: partial pivoting fixes row issues, and complete pivoting additionally controls columns, which is visible in smaller γ and flat relF/relR curves.

In the large-n trends, partial and complete pivoting keep relF and relR near round-off and concentrate γ with low means. Overall, the implementation achieves stable factorizations and accurate solutions with modes 1–2 across the tested cases. The data support the standard recommendation to use at least partial pivoting in practice, with complete pivoting offering the tightest control of element growth when needed.

10 Disclaimer

ChatGPT was used to help debug and develop the drivers. I used Dr. Gallivan's driver from Program 1 for Set 2. While I had to rely heavily on ChatGPT to develop the driver to reliable detect issues and plot usable figures. I made the attempt to comment on most lines to make sure I understood what each step was doing dispite me not being able to develop this on my own. Subroutines were largly developed by me but I have had communication with other student such as Scott, Yunxi and Roujin as well as others. ChatGPT was used to help orgainze content and develop the workflow since I found the program3 outline particularly unclear.

A Figures

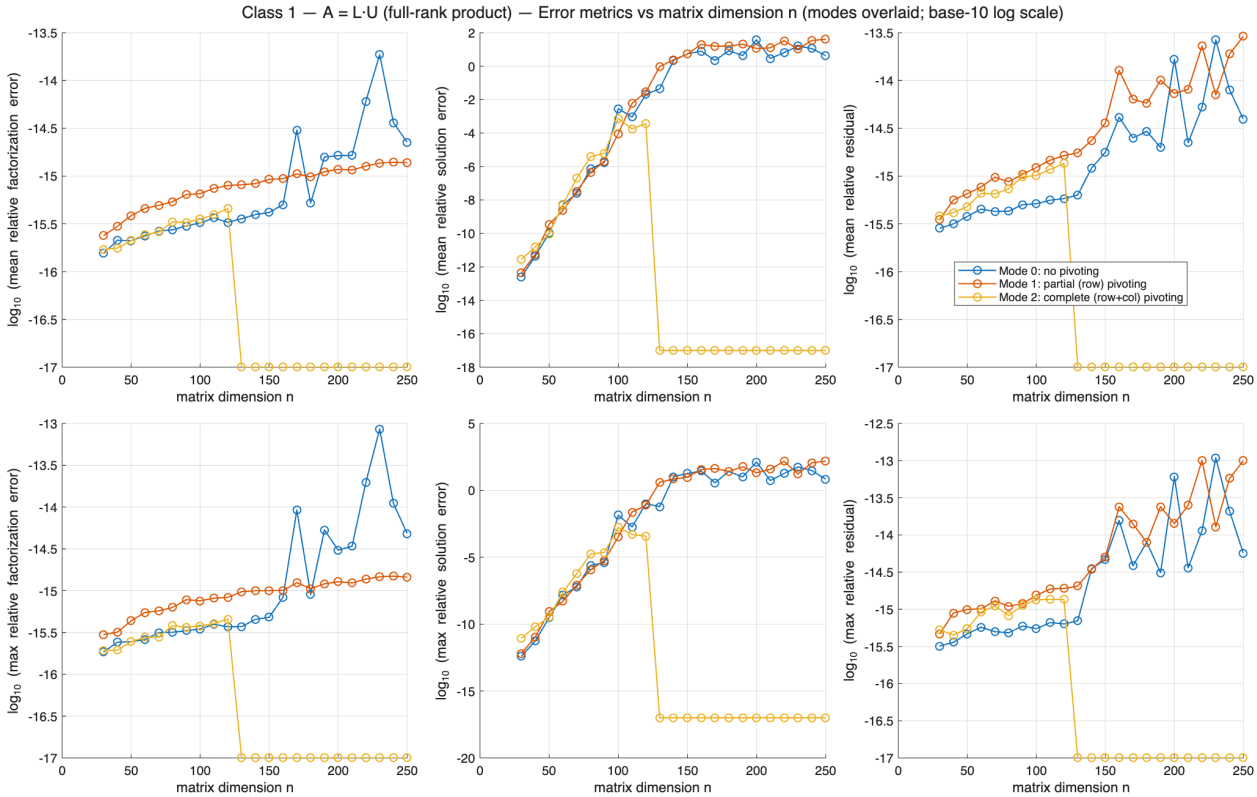

Figure 1: Dashboard: $\log_{10}$ mean/max errors vs. n (Class 1), modes overlaid.

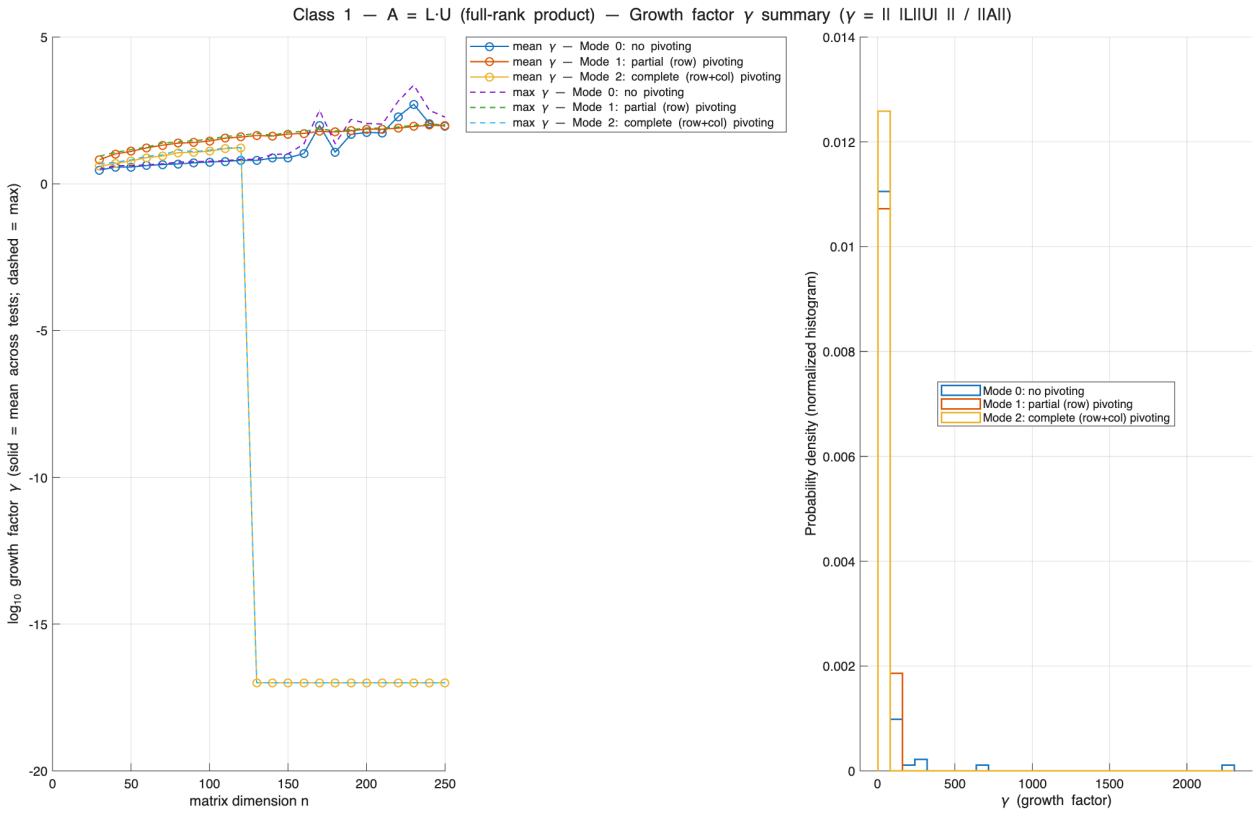

Figure 2: Growth factor summary for Class 1: mean/max γ vs. n (left) and overlaid γ histograms by mode (right).

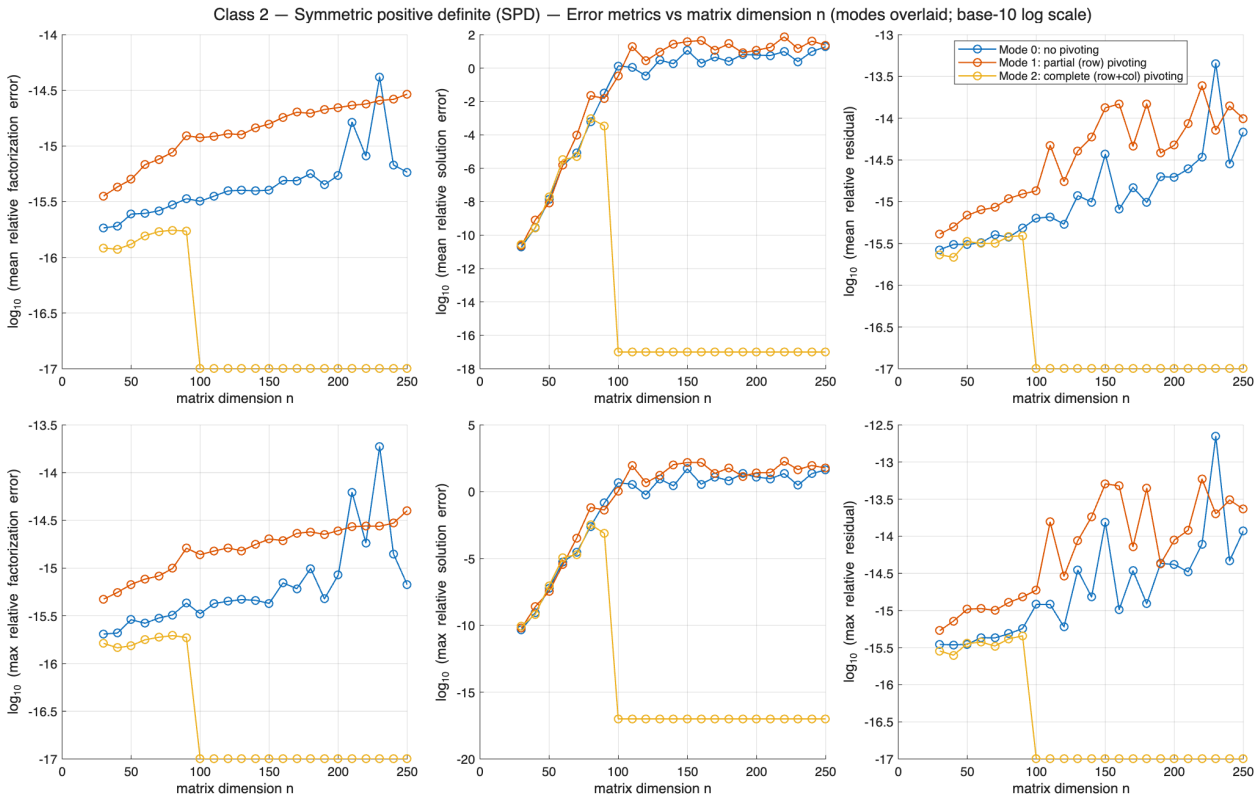

Figure 3: Dashboard: Class 2 (SPD).

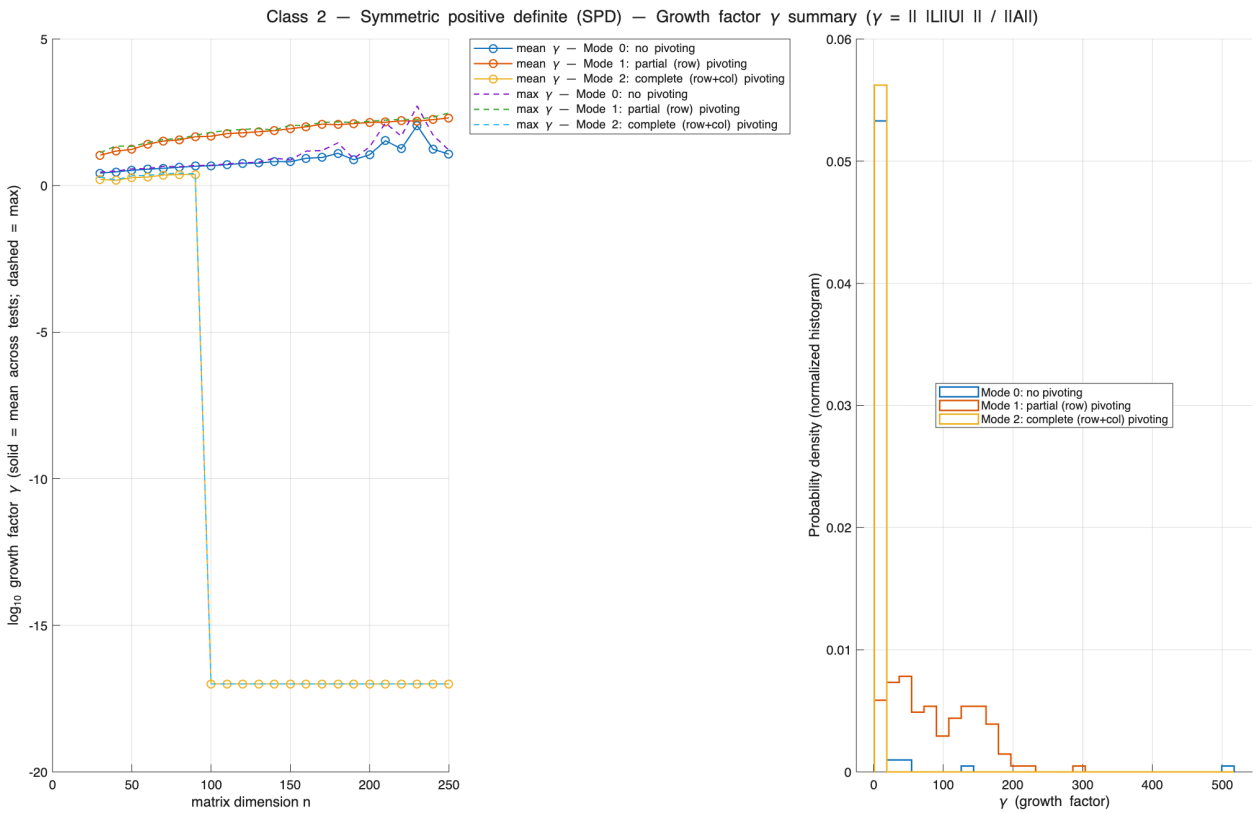

Figure 4: Growth factor summary: Class 2.

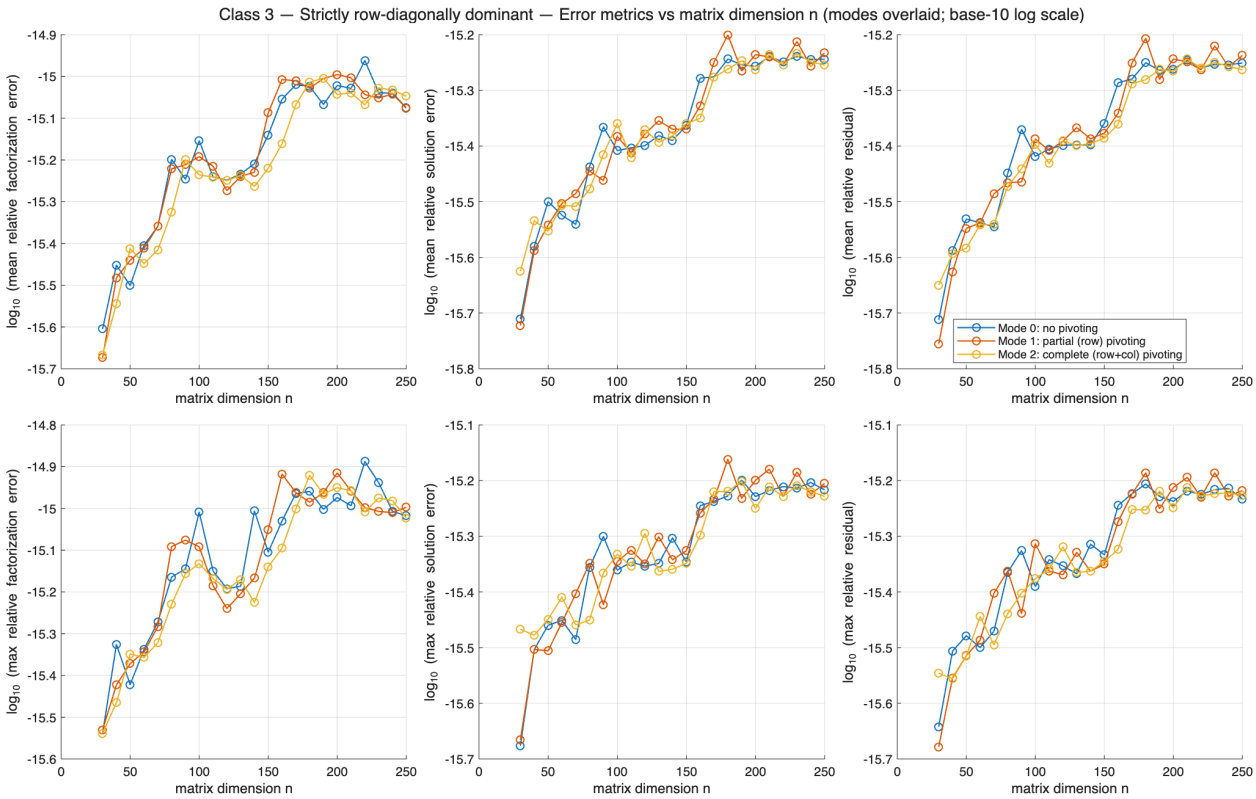

Figure 5: Dashboard: Class 3 (strict row-diagonal dominance).

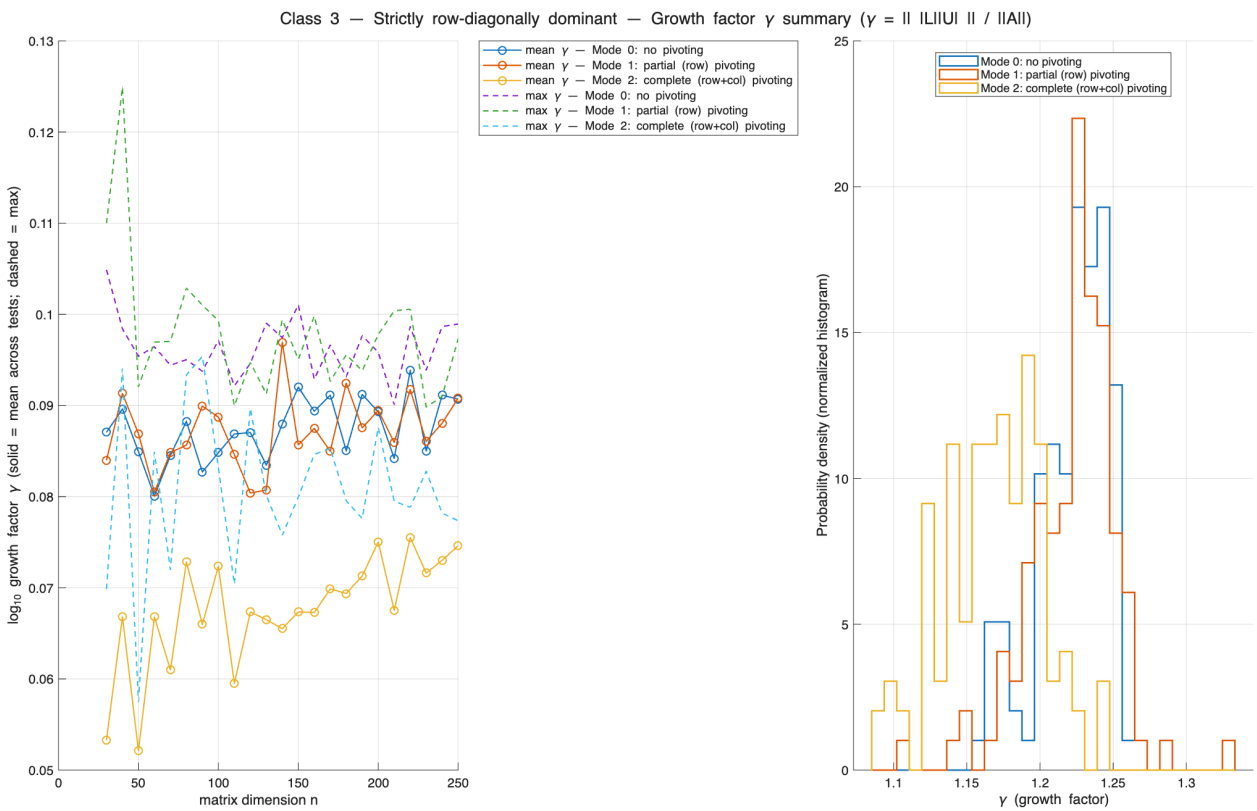

Figure 6: Growth factor summary: Class 3.

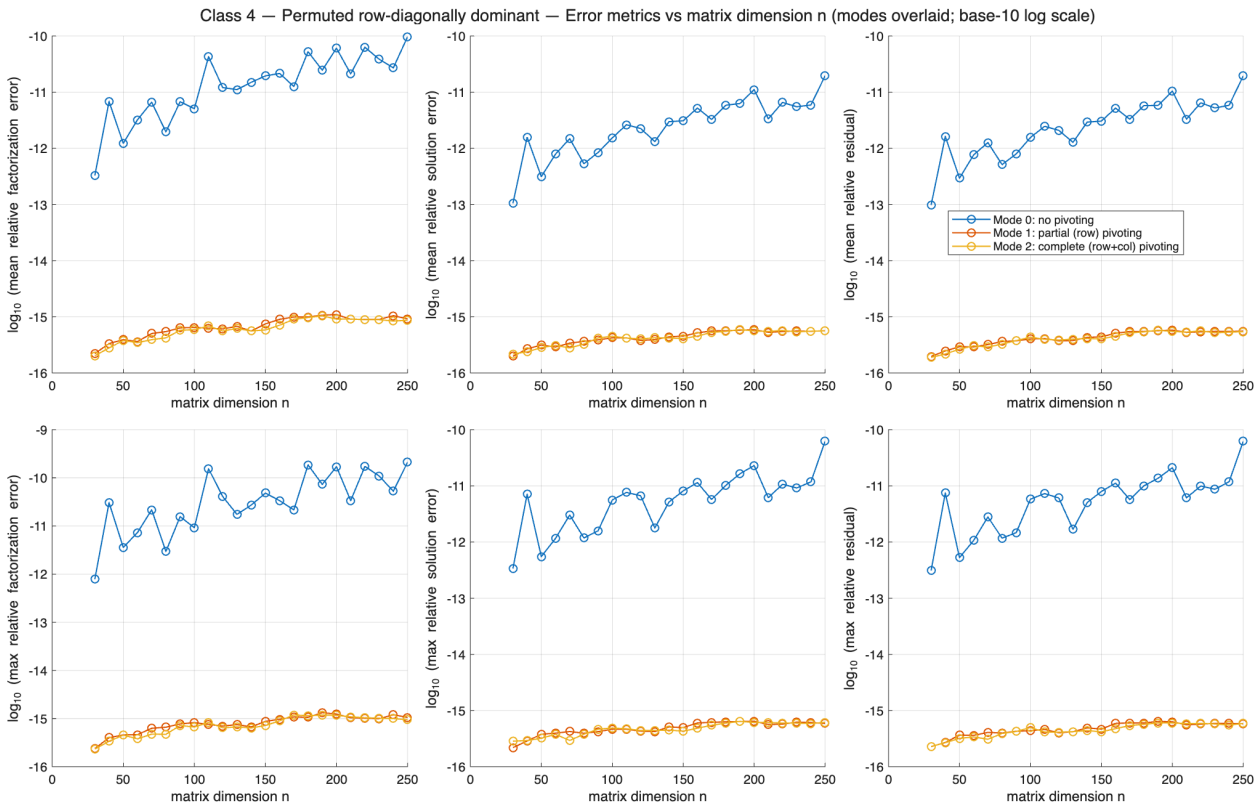

Figure 7: Dashboard: Class 4 (permuted diagonally dominant).

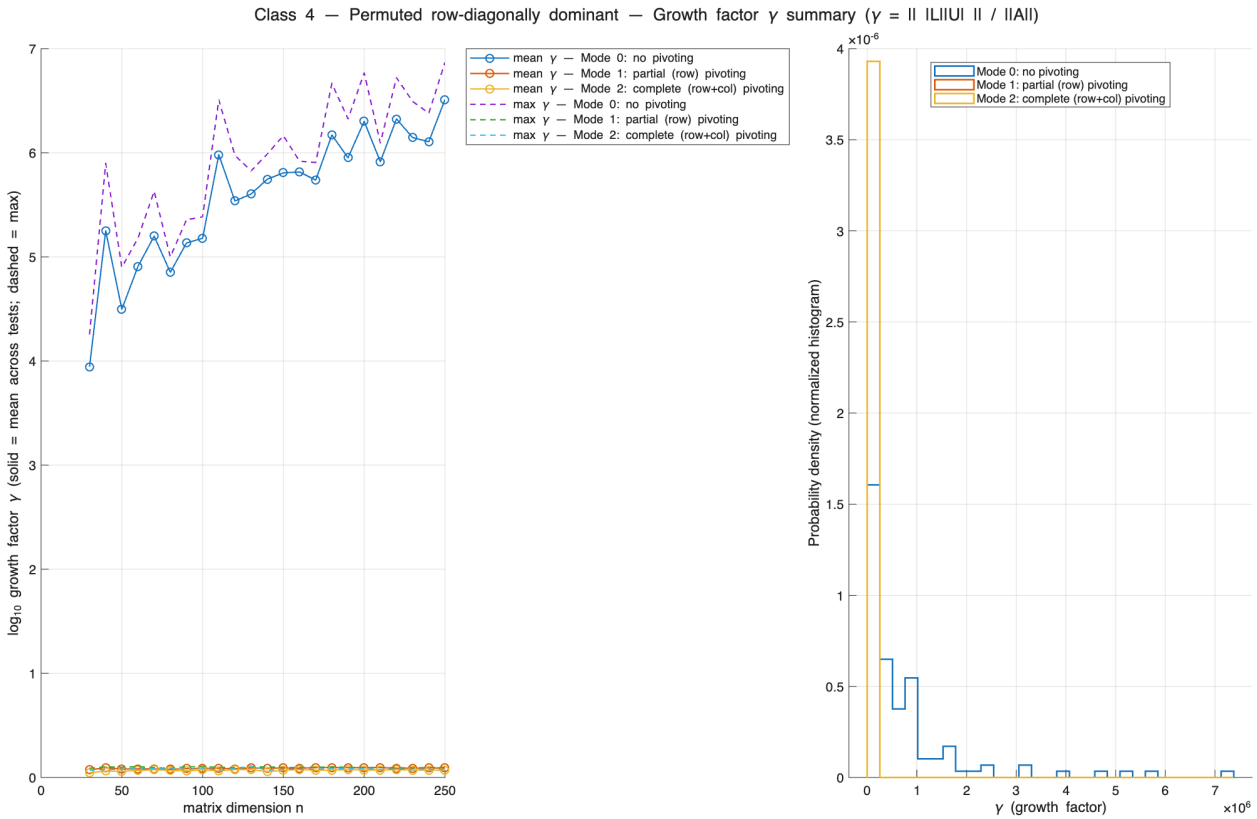

Figure 8: Growth factor summary: Class 4. The histogram's y–axis is *PDF* (probability density), so the area under each curve is 1.